

Digest: Why is Quantum Physics Based on Complex Numbers?

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1 Introduction

This document formally digests Felix M. Lev's paper on Galois Field Quantum Theory (GFQT). The theory substitutes the standard complex Hilbert space with a linear projective space over a finite field $GF(p^n)$.

2 Modular Representations and $GF(p^2)$

The paper proves that a finite field analog of the de Sitter algebra $so(1,4)$ requires a quadratic extension $GF(p^2)$ where $p \equiv 3 \pmod{4}$ to ensure that spinless irreducible representations are fully decomposable. This quadratic extension acts identically to the complex numbers $\mathbb{C}$, supplying the required imaginary root $i^2 = -1$.

3 $so(1,4)$ Algebra

The symmetry algebra generators M^{ab} satisfy:

$$[M^{ab}, M^{cd}] = -2i(\eta^{ac}M^{bd} + \eta^{bd}M^{ac} - \eta^{ad}M^{bc} - \eta^{bc}M^{ad}) \quad (1)$$

where $\eta = \text{diag}(1, -1, -1, -1, -1)$.

4 Modular $su(2)$ and Operators

The algebra includes the $su(2) \times su(2)$ operators $\mathbf{J}', \mathbf{J}''$ and R_{jk} . The M_{04} operator, analogous to the energy operator, is analyzed in the $(\mathbf{J}^2, J_3)$ basis, yielding an equidistant imaginary spectrum. This confirms that M_{04} is fully decomposable without introducing any inconsistencies in $GF(p^2)$.